

STARFALL WORKS

Wargame Counter Creator

User Manual - Version 0.44

Design individual counters. Build two-sided sets. Arrange printable sheets. Export production-ready PDFs.

Local browser application - no installation or server required

Prepared for Wargame Counter Creator v0.44

How to Use This Manual

This manual follows the normal production workflow: create counters, choose a layout and symbol, refine artwork and values, define front/back behavior, arrange sheets and export production files. It also explains bulk editing with CSV and the manufacturer-specific Superior POD workflow.

Version 0.44 includes all features through the searchable NATO / APP-6 symbol browser. References to Superior POD apply to the bundled 5/8 inch, 3/4 inch and 1 inch manufacturer templates.

What changed since the previous manual

- Large Ship template for 3/4 inch and 1 inch naval counters.
- Superior POD export and exact-slot planning for 5/8 inch, 3/4 inch and 1 inch sheets.
- CSV import for bulk editing existing counters by Counter Number.
- Selectable silhouette color for built-in and imported artwork.
- Automatic cleanup of opaque backgrounds in imported silhouettes.
- Damage explosion overlay behind the silhouette with a selectable color.
- NATO / APP-6 symbol support with compact search, category filtering and direct SIDC entry.
- The former Old School symbol group is no longer shown; older projects are migrated when loaded.

1. Getting Started

1.1 Launching the application

1. Extract the Wargame Counter Creator ZIP file to a folder on your computer.
2. Open the extracted folder.
3. Double-click index.html, or open it in Chrome, Edge, Firefox or Safari.
4. The application runs locally in the browser. No server or installation is required for the core application.

NATO / APP-6 rendering currently loads the milsymbol browser library from the internet. If you are offline, the rest of the application remains usable, but NATO symbols may not render until the library is available.

1.2 Main workspace

The application has two primary workspaces:

- Counter Designer - creates and edits individual counter designs and their reverse sides.
- Sheet Designer - arranges counters for general printing and Superior POD production.

The main toolbar includes project creation/saving/loading, CSV export/import, General Double-Sided PDF, Superior POD PDF and Browser Print / PDF functions.

1.3 Recommended file workflow

- Use project JSON as the master editable file.
- Use CSV as a bulk-editing interchange format, not as the only project backup.
- Use direct PDF export for production whenever possible.
- Keep the final project JSON beside the final PDF so the counter set can be revised later.

2. Counter Designer

2.1 Creating a counter

5. Add a new counter from the Counters panel.
6. Choose the template and physical counter size.
7. Enter the unit name, type and values that apply to the selected template.
8. Choose colors, symbol or imported artwork.
9. Set Quantity to the number of physical copies required.
10. If the piece is reversible, enable Two-sided and edit the back separately.

2.2 Physical counter sizes

Size	Decimal inches	Typical use
1/2 inch	0.500	Small traditional counters
5/8 inch	0.625	Common traditional size; Superior POD supported
3/4 inch	0.750	Larger units and ships; Superior POD supported
1 inch	1.000	Large counters and information-heavy pieces; Superior POD supported

Physical size is the source of truth for direct PDF output. For example, a 5/8 inch counter is 0.625 inches wide, or 45 PDF points at 72 points per inch.

2.3 Bleed and safe area

Bleed extends background artwork beyond the finished counter edge so minor cutting variation does not reveal white paper. The safe inset identifies an interior area where important text and values should remain.

- Bleed presets include None, 1/32 inch, 1/16 inch and 1/8 inch.
- Safe inset presets include 0.025 inch, 0.040 inch and 1/16 inch.
- Use Show guides in the Counter Designer to preview the boundaries.

3. Counter Templates

3.1 Classic

Classic is the traditional combat-unit layout with a unit name, unit type, centered symbol and three bottom values.

3.2 Six-Value

Six-Value adds optional top-left and top-right values while retaining three optional bottom values. The symbol occupies its own image zone and the piece name is placed in a separate text band.

- All five numeric positions may be left blank.
- Each numeric position has independent text color, highlight on/off and highlight color.

3.3 Information

Information counters are intended for turn markers, phases, weather, victory points, control markers, reminders and other non-unit pieces. They emphasize centered text and may include a built-in or imported image.

3.4 Large Ship (3/4 inch and 1 inch)

Large Ship is designed for larger naval counters. It is available at 3/4 inch and 1 inch and provides eight optional numeric positions around a central ship silhouette, plus country, ship name, letter and ship-type fields.

- Selecting Large Ship with a smaller size automatically raises the counter to 3/4 inch.
- All eight numeric positions support independent text color and highlight styling.
- Large Ship counters can be two-sided like the other templates.
- They can be exported through the 3/4 inch and 1 inch Superior POD workflows.

4. Symbols, Silhouettes and NATO / APP-6

4.1 Built-in silhouettes

The traditional silhouette library includes None, Infantry, Tank, Artillery, Truck, Fighter, Bomber, Destroyer, Cruiser, Battleship, Carrier, Submarine and Helicopter.

4.2 Silhouette color

Silhouette color is selectable per counter side. The default is black. The color applies to built-in silhouettes and to imported artwork after it is normalized into a mask.

4.3 Importing custom symbols

11. Import a PNG, JPG or SVG using the Imported symbol control.
12. Select the imported symbol for the current counter.
13. Choose the desired Silhouette color.
14. Use Remove Imported Symbol if the artwork is no longer needed.

Imported artwork is stored inside the project JSON. When an imported image has an opaque white, cream or gray background, the application estimates that background from the image corners and removes it before applying the silhouette color. Images that already contain true transparency preserve their alpha channel.

4.4 NATO / APP-6 symbol browser

Choose NATO / APP-6 Symbol from the symbol selector to expose the compact NATO browser. The browser is intended for standard military unit symbology and works alongside the traditional silhouette library.

- Search - type a term such as infantry, armor, artillery, engineer, reconnaissance, medical or supply.
- Category - filter the friendly-name catalog by Combat, Combat Support, Combat Service Support, Command & Control or Aviation.
- Matches - choose a matching symbol from the result list.
- SIDC - enter a supported Symbol Identification Code directly when the desired symbol is not in the friendly-name catalog.
- Show NATO frame - optionally include the NATO affiliation/frame around the symbol.

The friendly-name catalog is a convenience layer. Direct SIDC entry remains available for additional symbols supported by the APP-6 renderer.

4.5 Older Old School symbols

The separate Old School symbol group is no longer shown because those box-based marks overlap with NATO / APP-6 symbology. When an older project is loaded, Old School Infantry, Cavalry, Artillery and Armor are migrated to corresponding NATO symbols. Earlier aircraft entries are mapped to the closest traditional aircraft silhouette.

5. Styling, Fonts and Damage Effects

5.1 Colors

- Background - main counter field color.
- Border - finished counter edge color for non-POD output.
- Text - default label color.
- Silhouette - independent symbol color.
- Per-number colors - independent text and optional highlight colors for each number position.

5.2 Background stripes

Every template supports an optional vertical or horizontal stripe behind the artwork and text. The stripe has its own color and may be positioned Left/Center/Right for vertical stripes or Top/Center/Bottom for horizontal stripes.

5.3 Font size controls

- Label font size (%) controls names, unit types, Information text and Large Ship labels.
- Number font size (%) controls numeric values independently.
- Both controls range from 50% to 200% and are stored per face on two-sided counters.

5.4 Damage explosion overlay

Enable Damage explosion to place a starburst/explosion graphic behind the silhouette. Use Explosion color to choose the effect color. The silhouette remains in the foreground.

- The effect appears only when a symbol or silhouette is present.
- Front and back faces may use different damage settings.
- The effect is supported in on-screen preview, general PDF and Superior POD PDF output.
- CSV export/import includes Damage Explosion and Damage Explosion Color for bulk editing.

6. Two-Sided Counters

Two-sidedness is defined on the counter record so front and back remain paired through quantity expansion, sheet layout and export.

6.1 Creating a two-sided counter

15. Select the counter.
16. Turn on Two-sided counter.

17. Use Front and Back to switch between faces.
18. Optional: choose Copy Front to Back to use the front as the starting point.
19. Edit the back independently.

The front and back share physical size, bleed and safe-area geometry. Other content and visual properties may differ, including template, colors, symbol, values, fonts and damage explosion settings.

6.2 One-sided backs

A one-sided counter does not require a separate reverse design. Direct exporters create a background/stripe treatment on the back without front text, symbol or numeric values unless Repeat fronts on backs is selected in the Superior POD workflow.

7. Managing a Counter Set

7.1 Quantity

Quantity controls how many physical copies of a design are produced. One record with Quantity 8 becomes eight physical copies in sheet/PDF output. For a two-sided counter, all eight fronts remain paired with their eight backs.

7.2 Duplicate, delete and reorder

- Duplicate creates a copy of the selected design for quick variation.
- Delete removes the selected design.
- Move Up / Move Down establishes the authoritative automatic production order.
- Quantity expansion happens after ordering, so identical copies remain together.

7.3 Save and load projects

Save Project writes a JSON file containing counter designs, imported symbols, two-sided data and Superior POD manual layouts. Use project JSON as the authoritative editable project file.

8. CSV Export and Import

8.1 Export CSV

Export CSV writes one row per counter design rather than one row per physical copy. The file is intended for documentation and bulk editing in Excel, Google Sheets or another spreadsheet tool.

- Counter Number and Quantity
- Template and Counter Size
- Names, unit types, Information text and symbols
- Background, stripe, border, text and silhouette colors
- Font-size percentages
- Numeric values, text colors and highlights
- Large Ship fields
- Damage explosion settings
- NATO SIDC and NATO Frame

8.2 Import CSV

Import CSV updates existing counters by Counter Number. It does not create duplicate counters.

20. Save the project JSON before making a large bulk change.
21. Export CSV from the application.
22. Open the CSV in a spreadsheet.
23. Edit the desired values.
24. Save/export the sheet as CSV.
25. Use Import CSV and select the edited file.

Only columns present in the imported CSV are changed. This allows a very small CSV to perform a focused bulk edit.

8.3 Example: change all counters from 5/8 inch to 3/4 inch

A two-column CSV can change only counter size:

Counter Number	Counter Size (in)
1	0.75
2	0.75
3	0.75

Fraction forms such as 5/8 and 3/4 are also accepted. When size changes on a two-sided counter, the back is kept at the same physical size so front/back registration remains correct.

8.4 CSV cautions

- Counter Number determines which existing design is updated.
- Blank text/stat cells in a column that is present are treated as intentional blank values.
- Use valid color values such as #112233.
- Large Ship counters are kept at 3/4 inch or larger.
- Keep a JSON backup before large spreadsheet changes.

9. Sheet Designer

The Sheet Designer controls general paper layout and the Superior POD workflow.

9.1 General sheet settings

- Paper size: US Letter or A4.
- Orientation: Portrait or Landscape.
- Margins and gutter spacing.
- Crop marks and guides.

Use Refresh Layout if the preview does not immediately reflect a changed sheet setting.

10. General Double-Sided PDF

General Double-Sided PDF is the preferred direct export when you are not using the Superior POD manufacturer format. It creates the PDF directly instead of relying on the browser print dialog.

10.1 Export behavior

- Uses the selected Letter/A4 size and orientation.
- Honors margins and gutters.
- Uses counter-list order and expands quantities.
- Supports mixed counter sizes.
- Creates each front page followed by its matching back page.
- Mirrors the back page horizontally for duplex registration.
- Uses true designed backs for two-sided counters.
- Uses background-only treatment for one-sided backs.
- Writes true physical PDF page dimensions while using high-resolution internal rasterization.

10.2 Printing at true size

Print direct PDFs at Actual Size or 100%. Do not use Fit to page, Shrink oversized pages or similar scaling options. Measure one known counter before committing to a large print run.

11. Superior POD Production

11.1 Supported manufacturer templates

Counter size	Front positions per sheet	Planner / export
5/8 inch	176	Exact manufacturer geometry
3/4 inch	126	Exact manufacturer geometry
1 inch	80	Exact manufacturer geometry

The ZIP includes reference copies of all three manufacturer template PDFs in the templates folder.

Exported production PDFs use coded manufacturer geometry and do not print the colored template guide artwork.

11.2 Choosing the POD size

In Sheet Designer, choose the Superior POD counter size. Automatic and manual planning then use only counters whose physical size matches that selected POD template.

11.3 Automatic layout

Automatic counter order fills manufacturer positions using the current counter-list order and quantity. It is the quickest production method when exact slot placement is not important.

11.4 Manual exact-slot planner

Manual slot layout lets you decide which counter occupies each manufacturer front position. Each POD size has its own saved manual layout.

- Fill from Counter Order - seed slots from the current counter-list order.
- Group by Color - create a starting arrangement grouped by background/stripe treatment.
- Add, clear or remove POD sheets.
- Click a front slot and use Selected slot to assign or replace a counter.
- Clear Slot empties the selected front position.
- Quantity summary shows assigned versus required copies and flags missing/over-assigned counters.

- Blank slots are supported.

The linked reverse position is calculated automatically from the manufacturer template. For 5/8 inch sheets the fronts/back are side-by-side; for the 3/4 inch and 1 inch templates the matching reverse positions are in the lower half of the portrait sheet.

11.5 Back behavior

- Two-sided counters use their independently designed back face.
- One-sided counters use a background/stripe-only reverse.
- Repeat fronts on backs overrides the normal reverse behavior when required.

11.6 POD bleed and borders

Superior POD output extends counter background colors into bleed/gap areas. Like-colored neighbors can form a continuous color field; different colors meet at the midpoint of the shared gap. POD output suppresses dark counter-border/divider strokes so the manufacturer cut geometry is not contaminated by printed black separation lines.

11.7 Large Ship and POD

Large Ship counters are intended for 3/4 inch and 1 inch production and are eligible for those two Superior POD formats. They are not eligible for the 5/8 inch POD template.

12. Browser Print / PDF

Browser Print / PDF is retained for convenience. Browser print dialogs vary and may rescale the page unexpectedly. For repeatable production output, prefer General Double-Sided PDF or Superior POD PDF.

13. Troubleshooting

Problem	What to check
Back button does not let me edit the reverse	Confirm Two-sided counter is enabled first.
Imported silhouette appears as a black rectangle	Use v0.40 or later. Re-load the project so the imported image is normalized; opaque backgrounds are removed automatically.
NATO symbol does not appear	Confirm the symbol is set to NATO / APP-6 and that the milsymbol renderer loaded. NATO rendering currently requires internet access.
NATO search does not contain the symbol I need	Enter its SIDC directly in the SIDC field.
PDF counter size is wrong on paper	Print at Actual Size / 100% and disable fit/shrink scaling.
POD export says no eligible counters	Confirm the selected Superior POD size matches the counters you are exporting.
Manual POD layout reports missing counters	Assign all required quantities or confirm that you intentionally want to export only assigned pieces.
CSV import skipped a row	Check that Counter Number is valid and refers to

	an existing counter.
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14. Recommended Production Workflow

26. Create one design for each distinct counter type.
27. Choose physical size before fine-tuning typography.
28. Choose the template and symbol system: traditional silhouette, imported artwork or NATO / APP-6.
29. Set colors, stripe, silhouette color and optional damage explosion.
30. Set label and number font sizes while viewing the counter at high zoom.
31. Use Quantity rather than duplicating identical counters.
32. For reversible pieces, enable Two-sided and complete the back face.
33. Use the counter-list arrows to establish automatic production order.
34. Save the project JSON.
35. If making a bulk change, Export CSV, edit it and Import CSV, then verify representative counters.
36. For home/general printing, configure Sheet Designer and use General Double-Sided PDF.
37. For Superior POD, select the matching 5/8, 3/4 or 1 inch template and choose automatic or manual placement.
38. If optimizing color/bleed neighborhoods, use Group by Color and manually refine slot placement.
39. Export the PDF and inspect it at 100%.
40. Verify one physical dimension with a ruler before production.
41. Archive the final project JSON, CSV if used and production PDF together.

15. Quick Reference

Task	Where to do it
Create or edit a counter	Counter Designer
Choose front/back	Two-sided counter + Front / Back controls
Bulk-edit size or colors	Export CSV -> spreadsheet -> Import CSV
Add a custom silhouette	Imported symbol control
Use NATO symbol	Silhouette / Symbol -> NATO / APP-6 Symbol
Mark unit damaged	Damage explosion + Explosion color
Set copies	Quantity
Control automatic order	Move Up / Move Down in counter list
General production PDF	General Double-Sided PDF
Superior POD production	Sheet Designer -> Superior POD size/layout -> Superior POD PDF
Exact POD placement	Manual slot layout

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